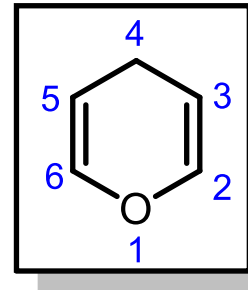
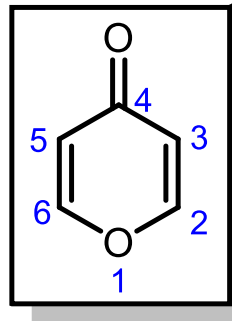


5.1.7. 4*H*-Pyran und Derivate

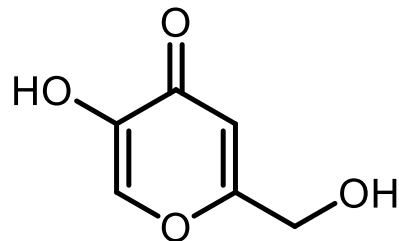
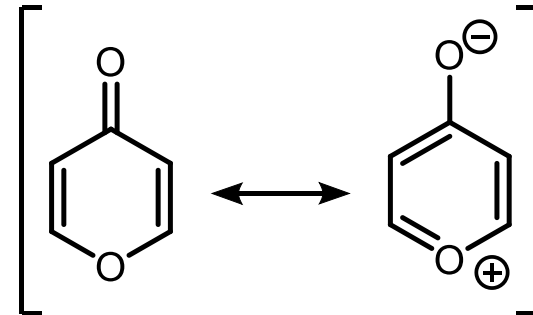
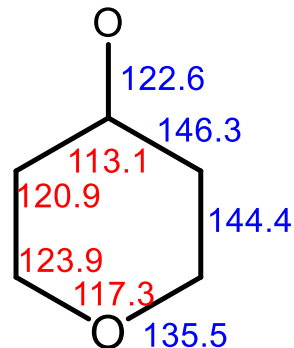
5.1.7.1. Stammsystem



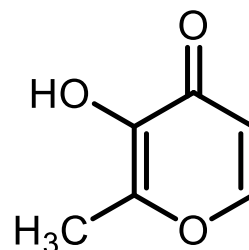
5.1.7.2. 4*H*-Pyran-4-on



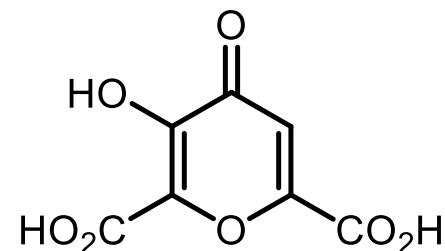
Bindungslängen [pm]
Bindungswinkel [°]



Kojisäure
(aus Zucker durch Bakterien)

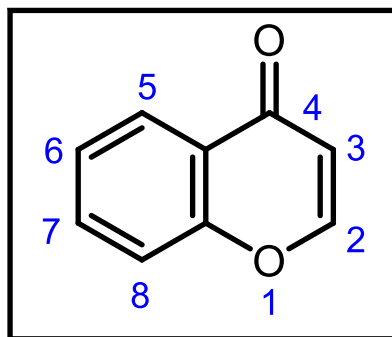


Maltol
(aus Lärchenrinde)

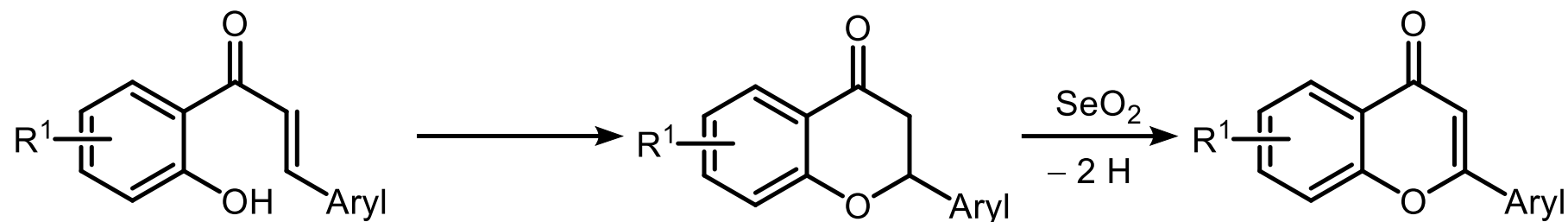
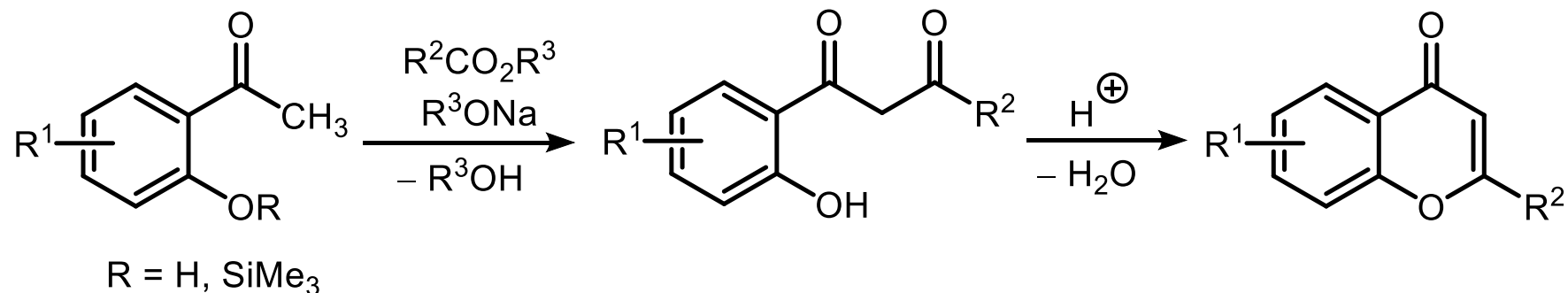


Mekonsäure
(häufig Salze der Opiate)

5.1.7.3. 4*H*-Chromen-4-on (Chromon)

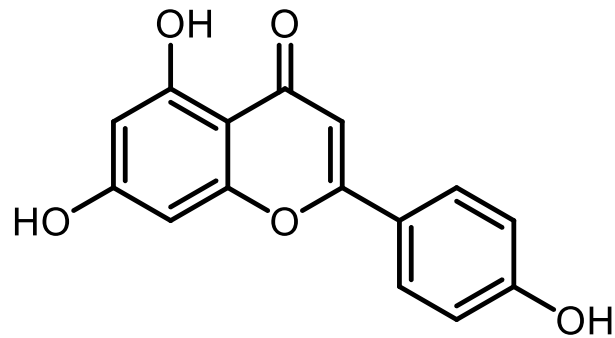


Chromon-Synthesen (ausgehend von *o*-Hydroxyacetophenonen)

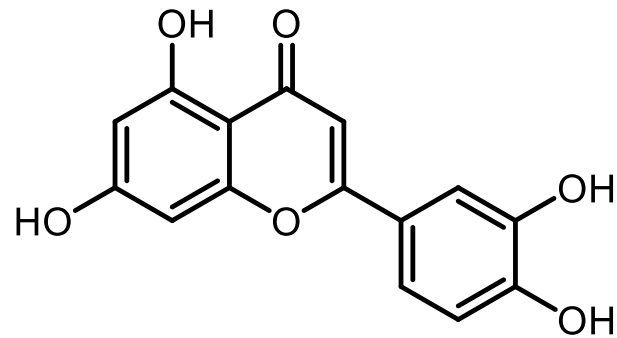


Chalkon
(durch Aldolkondensation)
Heterocyclen

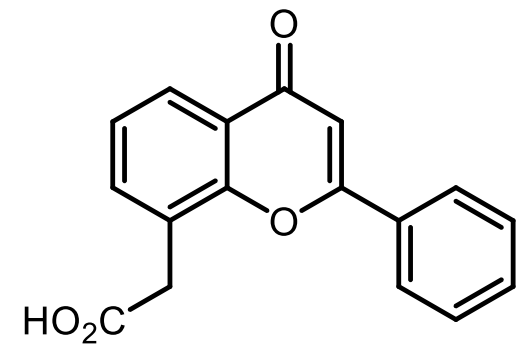
5.1.7.3. 4H-Chromen-4-on (Chromon)



Apigenin

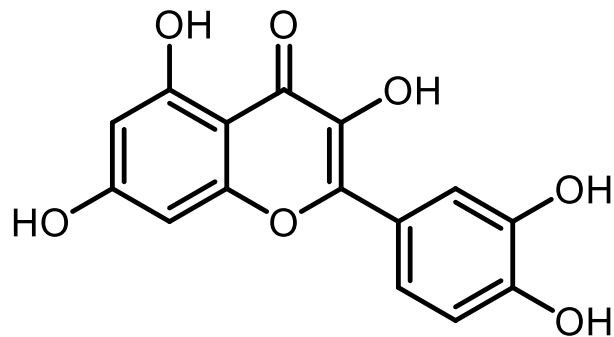


Luteolin

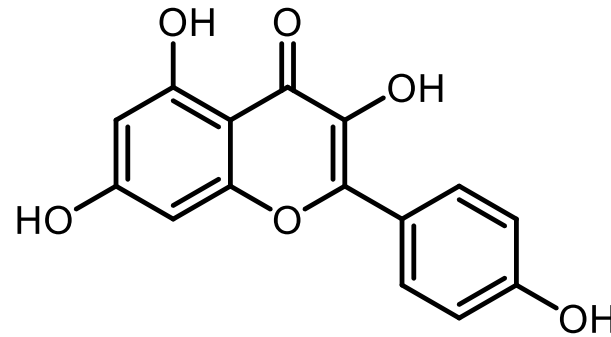


Flavon-8-essigsäure
(cytotoxisch, antitumor)

Flavonole

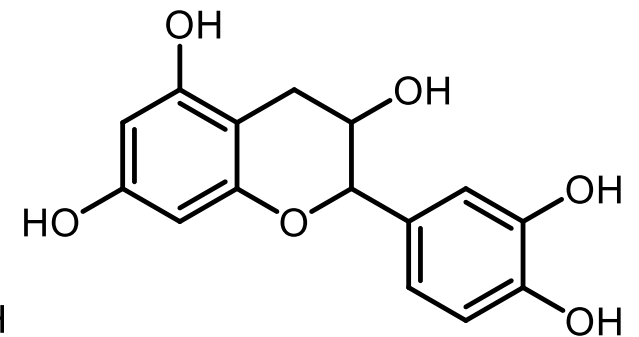


Quercetin



Kämpferol

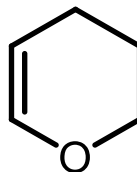
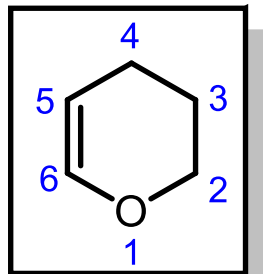
Flavanole



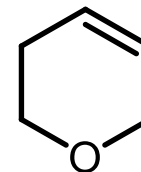
Catechin

5.1.8. Di- und Tetrahydropyrane und Derivate

5.1.8.1. 3,4-Dihydro-2*H*-pyran



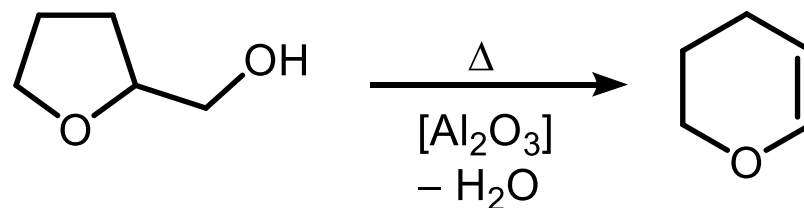
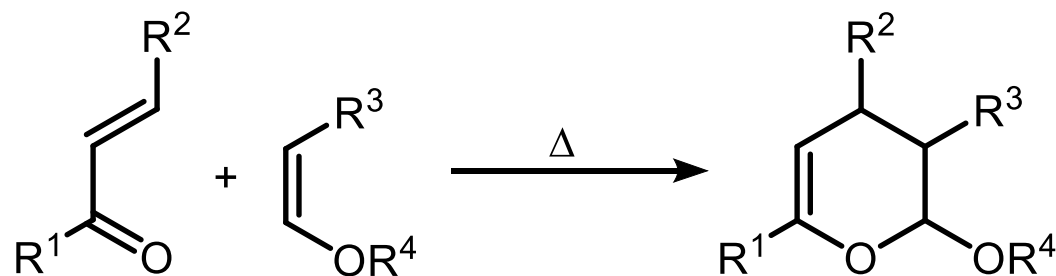
3,4-Dihydro-2*H*-pyran



5,6-Dihydro-2*H*-pyran

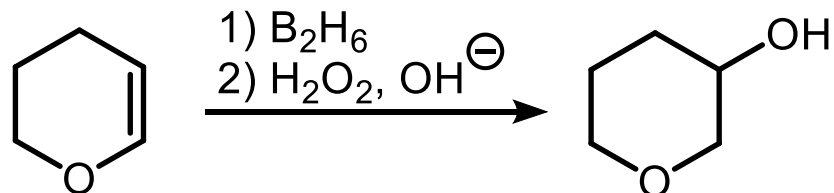
Synthese

91

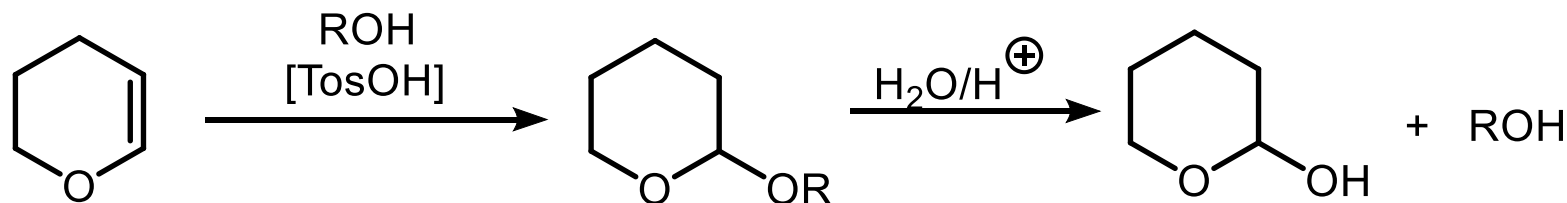


Reaktionen

92



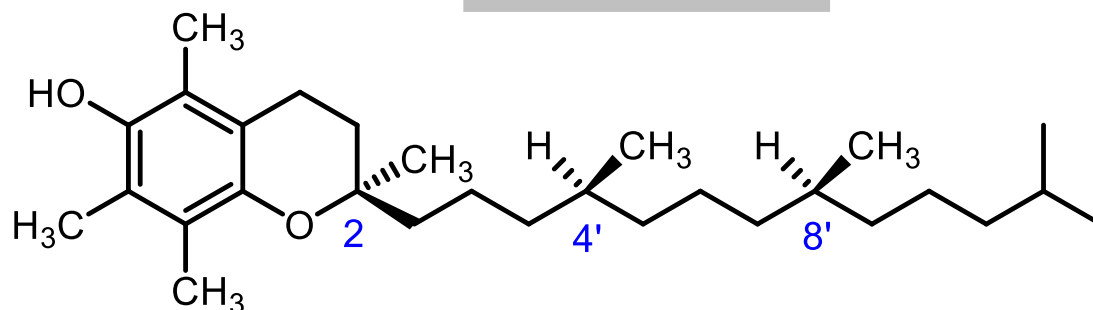
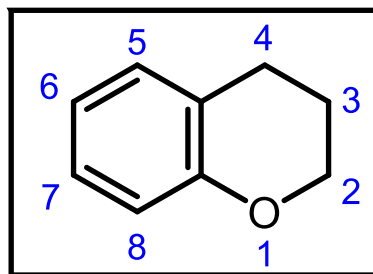
93



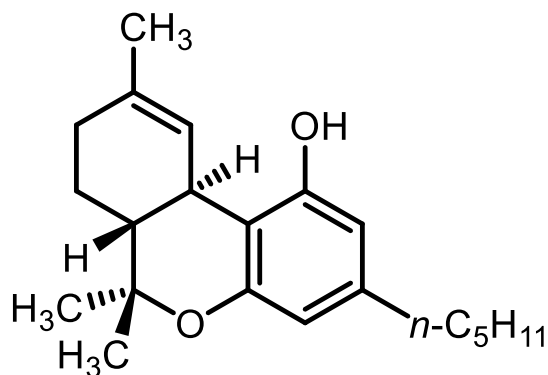
Bedeutung von DHP (3,4-Dihydro-2H-pyran):

OH-Schutzgruppe THP (Tetrahydropyran); allerdings Diastereomerenbildung bei chiralen Alkoholen

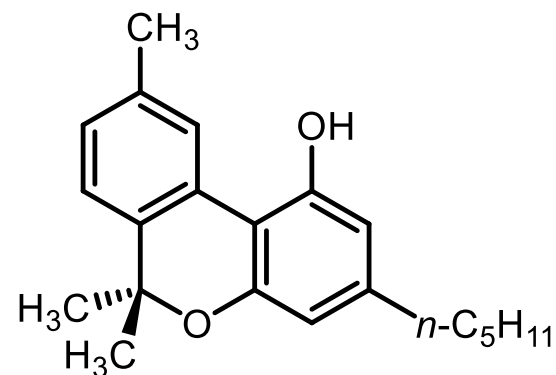
5.1.8.2. Chroman (Dihydrobenzopyran)



α -Tocopherol (Vitamin E)



Δ^9 -*trans*-Tetrahydrocannabinol (THC)

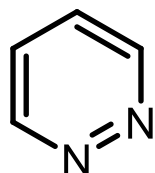


Cannabinol (CBN)

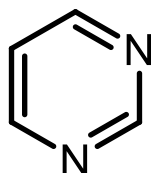
5.2. Sechsring-Heterocyclen mit zwei Heteroatomen

5.2.1. Diazine

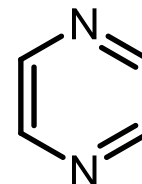
89



Pyridazin
 pK_S 2.3



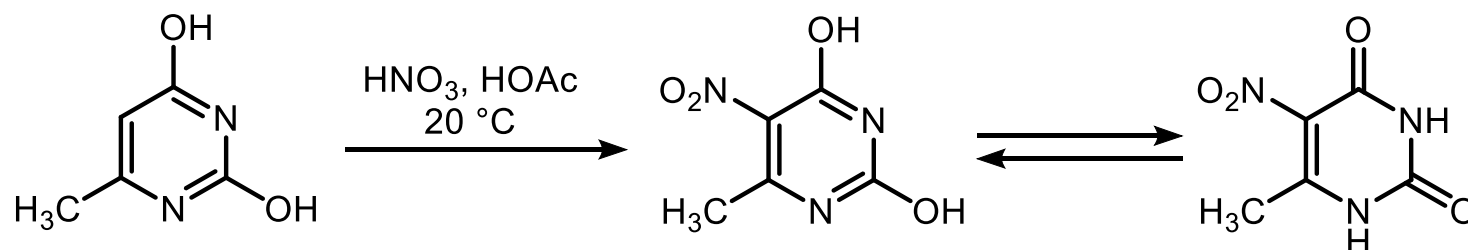
Pyrimidin
 pK_S 1.3



Pyrazin
 pK_S 0.5

5.2.1.1. Pyrimidin

S_EAr : geringe Reaktivität (ähnlich 1,3-Dinitrobenzol, 3-Nitropyridin) steigerbar durch elektronenliefernde Substituenten. Nucleophile Substitutionen besonders bei 2- oder 4-Halopyrimidinen.

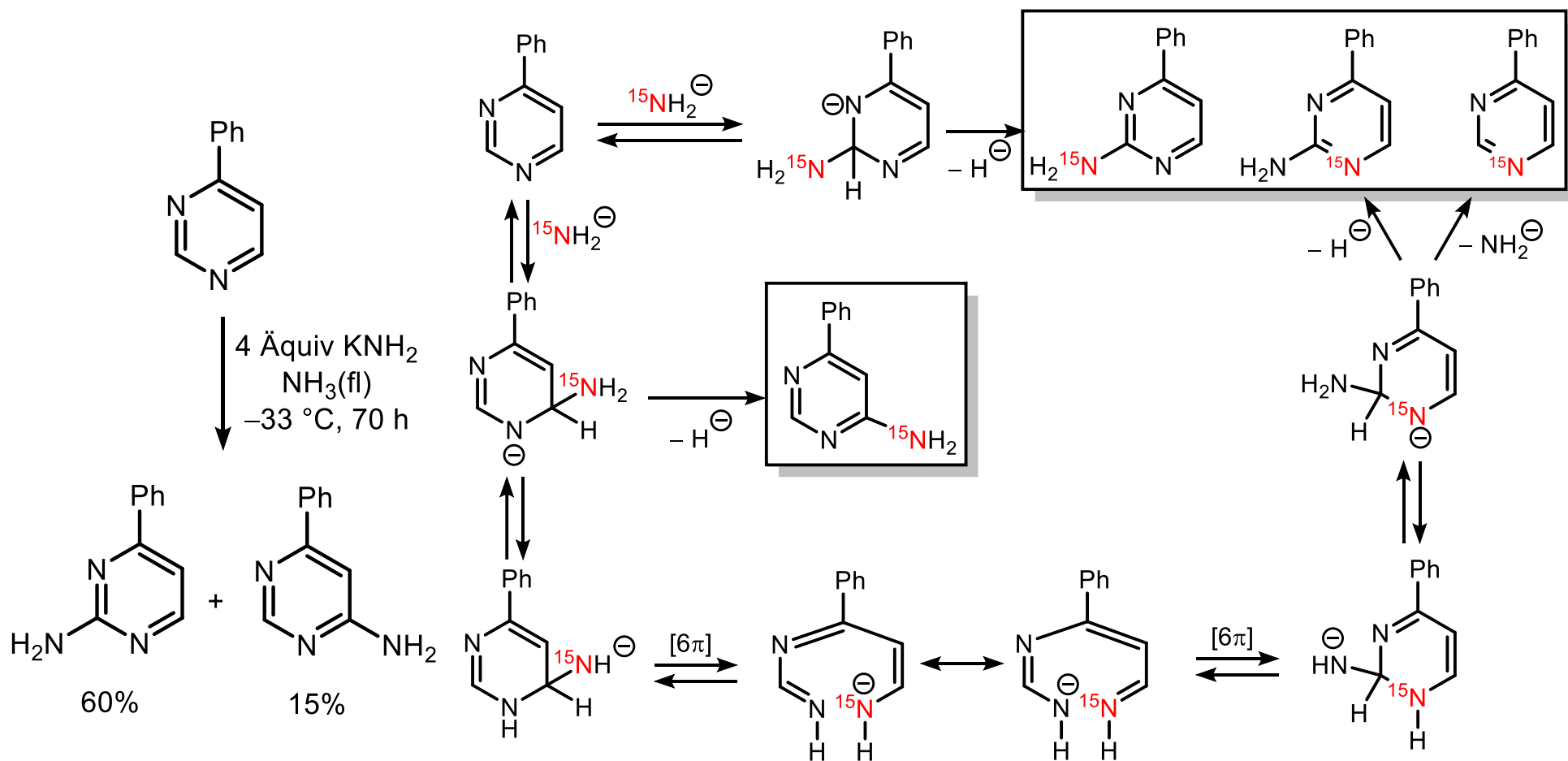


Nitrierung
Nitrosierung
Aminomethylierung
Azokupplung

Reaktivität wie Benzol

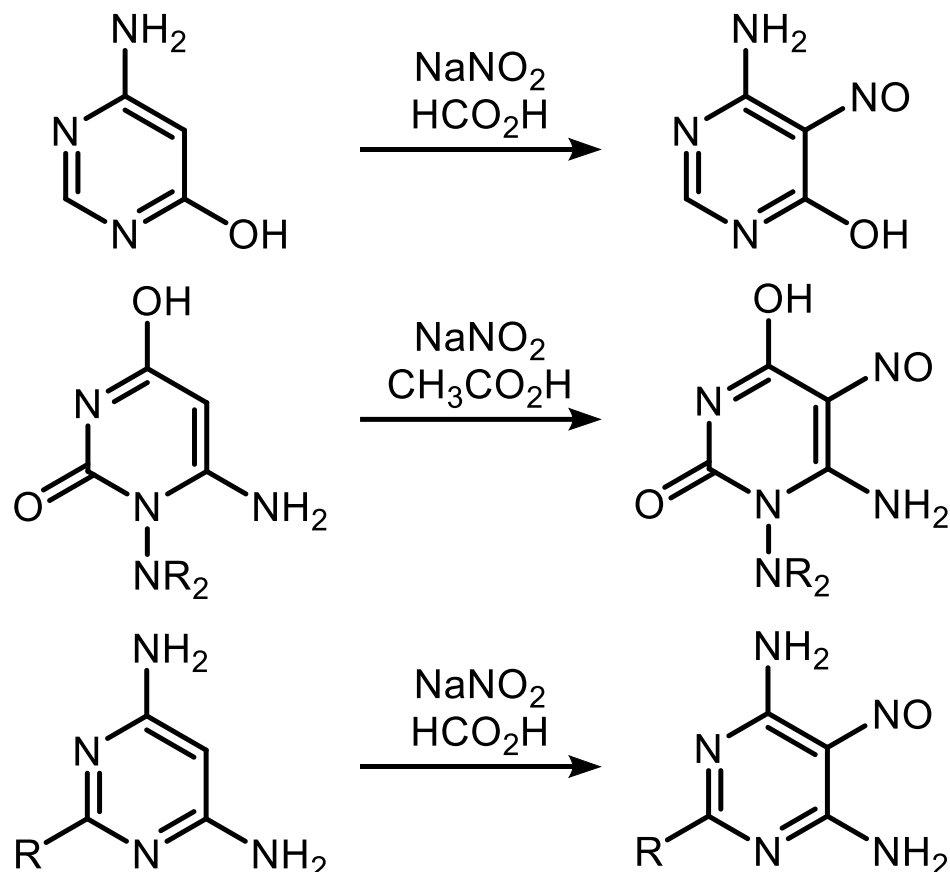
S_NAr bevorzugt an C4 und C6 (z.B. Organometallverbindungen)

Tschitschibabin-Reaktion von Pyrimidin

van der Plas, *J. Org. Chem.* **1979**, 4678.

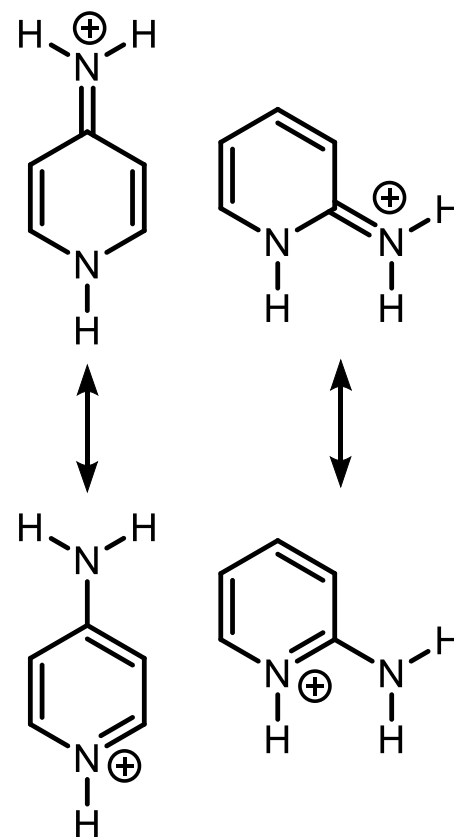
5.2.1.1. Pyrimidin

Ring-Nitrosierung in Konkurrenz zur Diazotierung

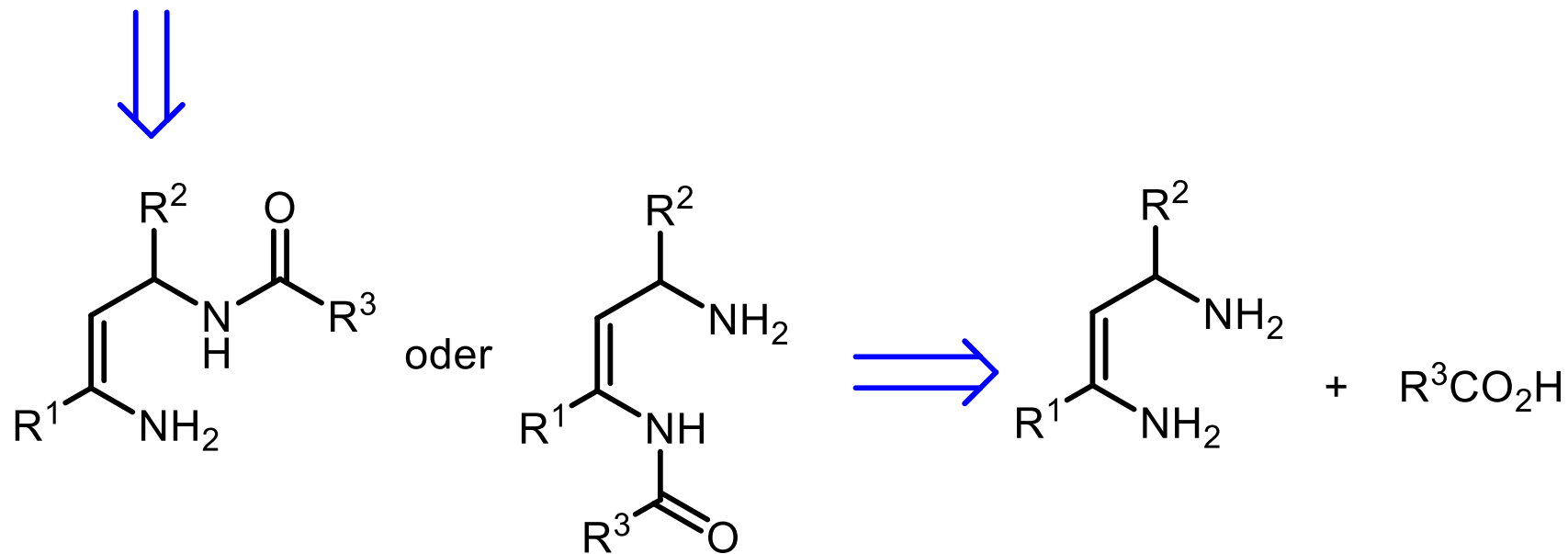
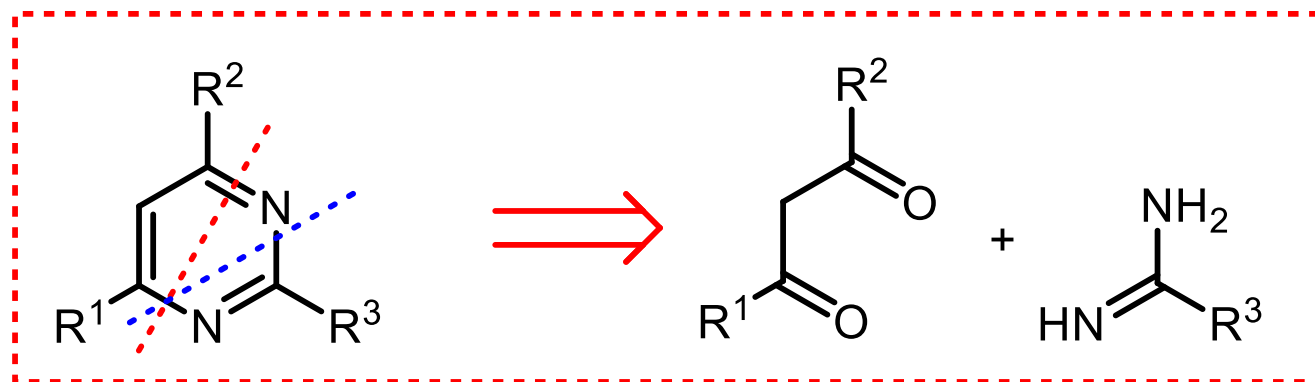


Butler, *Chem. Rev.* **1975**, 241.

Diazotierung ist immer dann gegenüber der Ring-Nitrosierung erschwert, wenn sich die Aminogruppe in 2- oder 4-Position zu einem Ring-Stickstoff befindet.

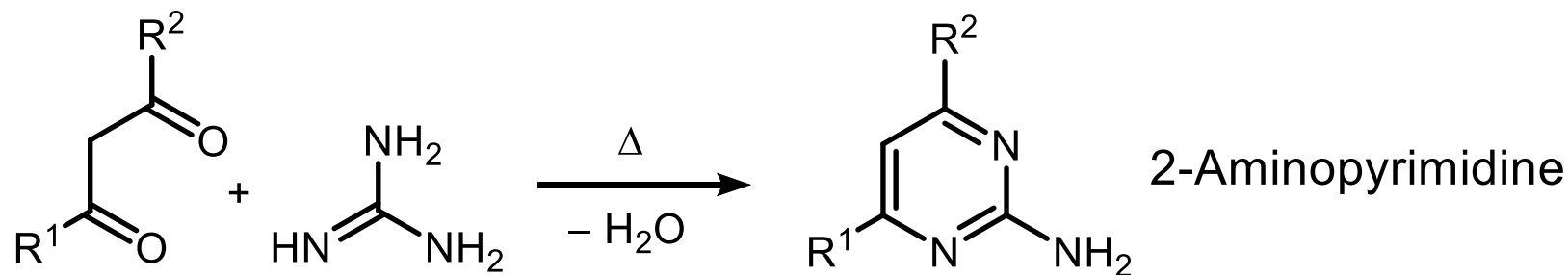
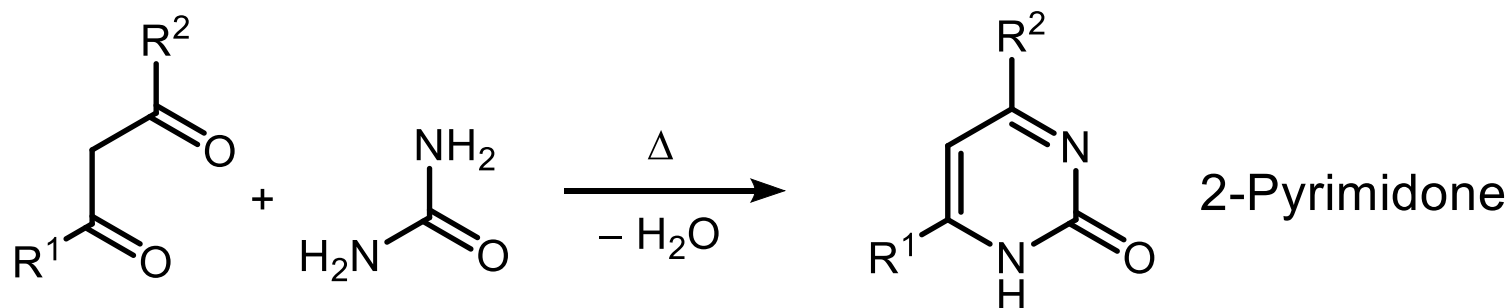
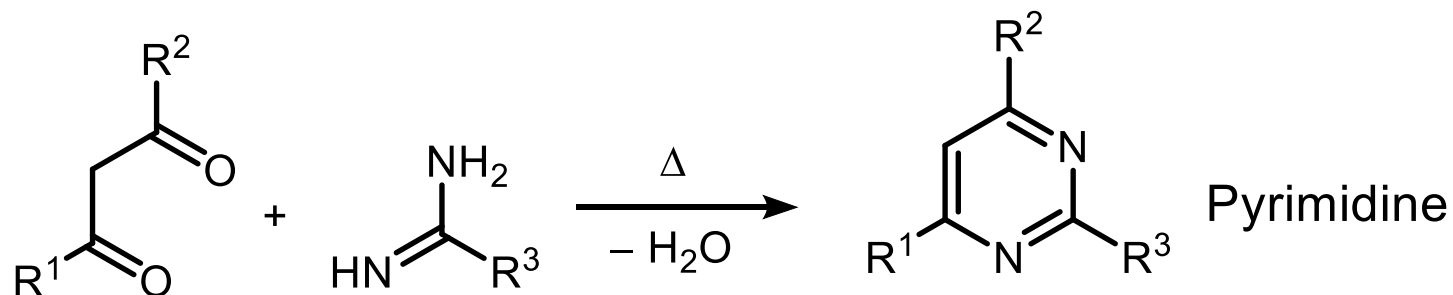


5.2.1.1. Pyrimidin Retrosynthese

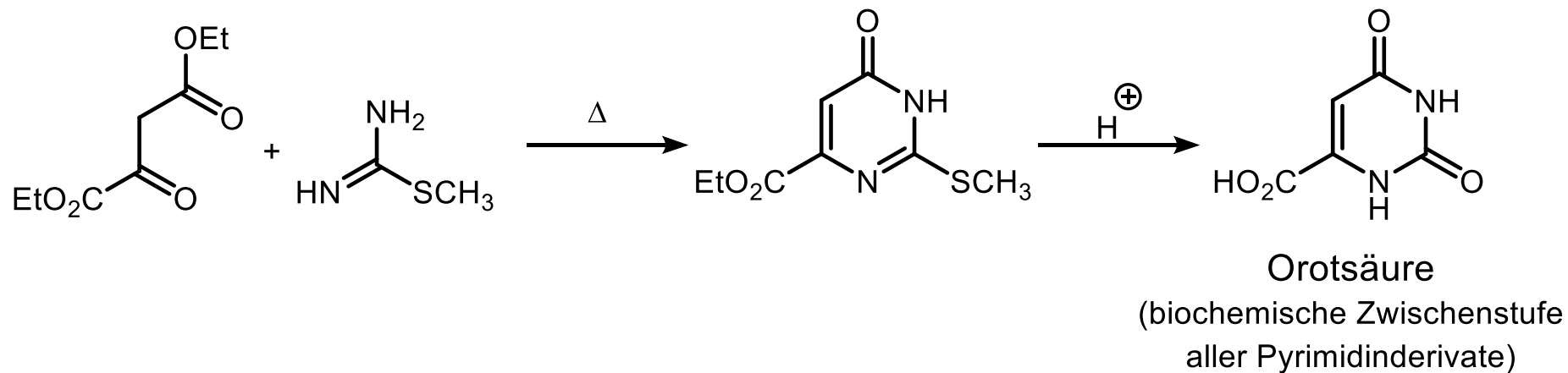
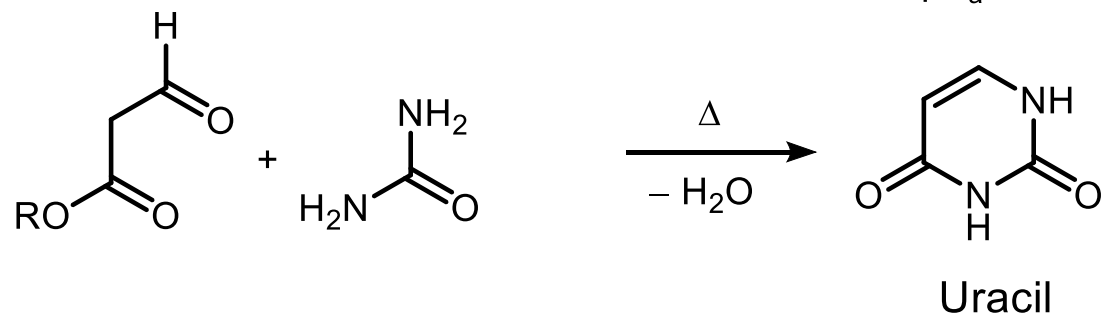
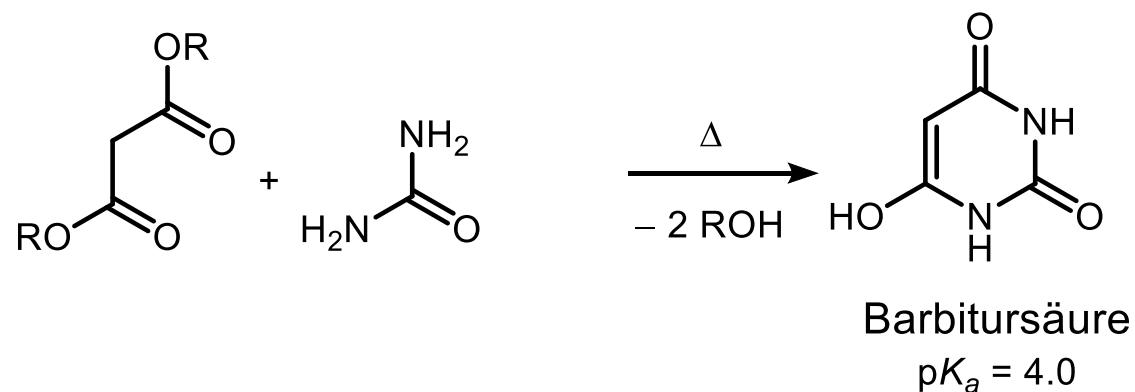


5.2.1.1. Pyrimidin Synthesen

1) Pinner-Synthese

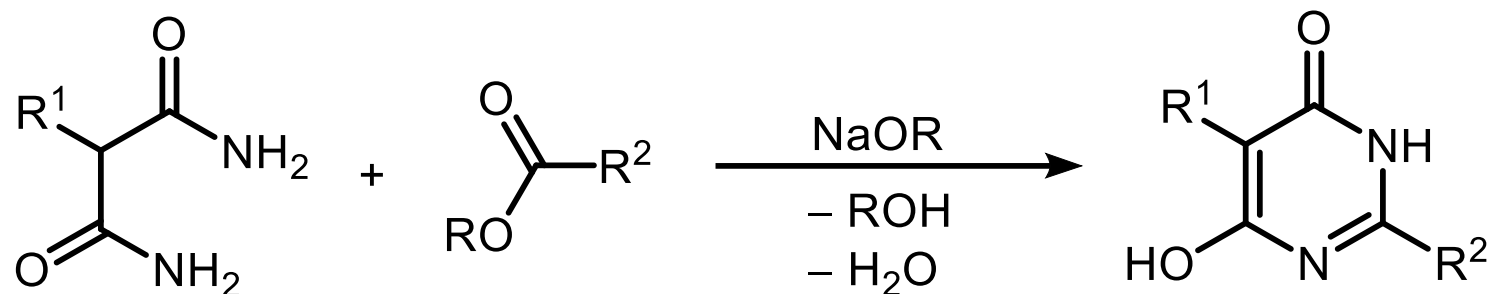


5.2.1.1. Pyrimidin Pinner-Synthese

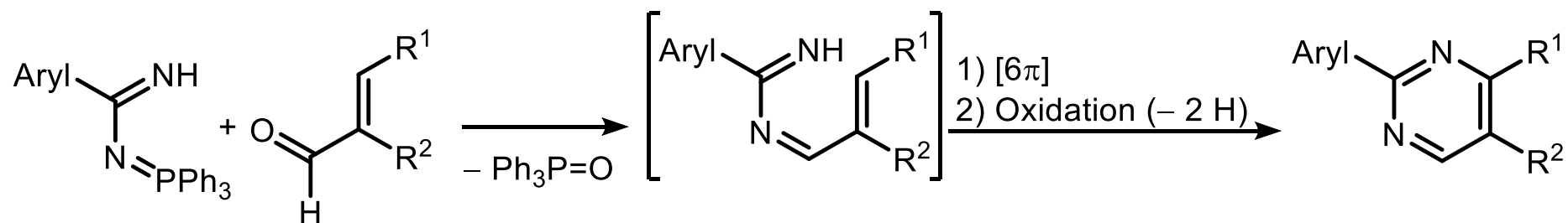


5.2.1.1. Pyrimidin Synthesen

2) Remfry-Hull-Synthese

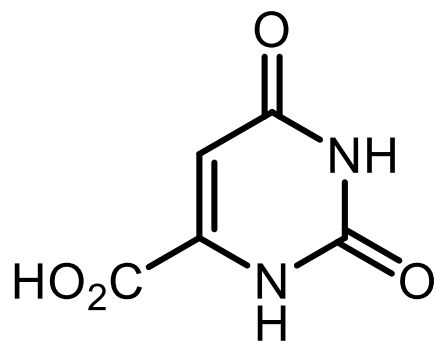


3) Über Aza-Wittig Reaktion



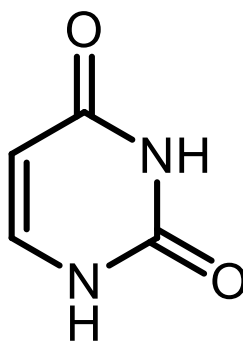
5.2.1.1. Pyrimidin

Vorkommen und Bedeutung



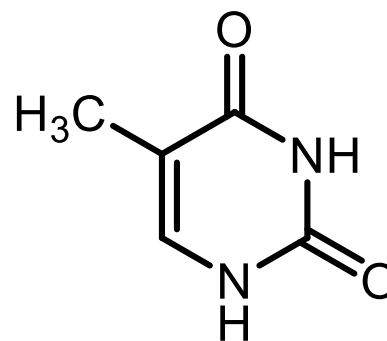
Orotsäure

(Biosynthesebaustein)

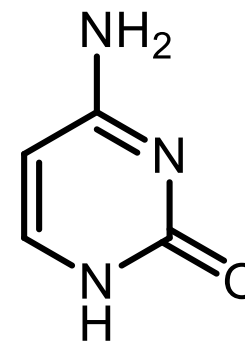


Uracil

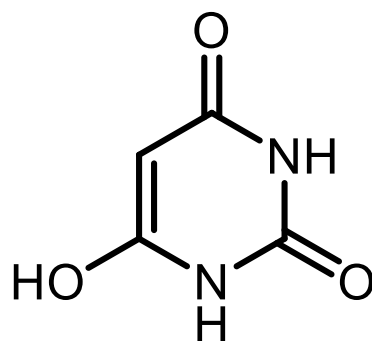
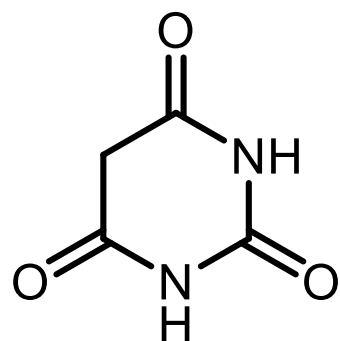
(Pyrimidinbasen in Nucleinsäuren)



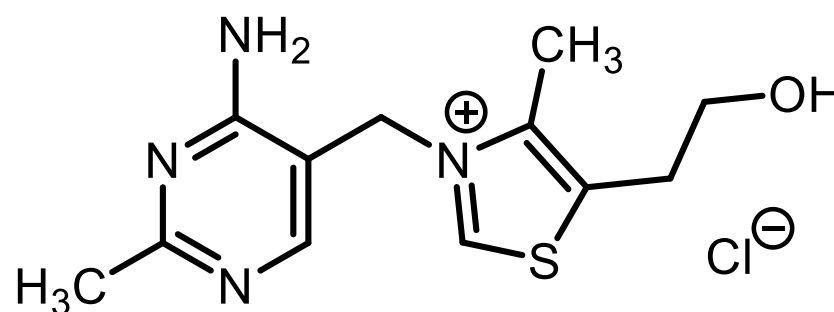
Thymin



Cytosin



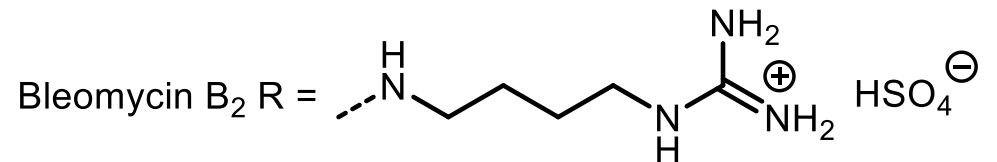
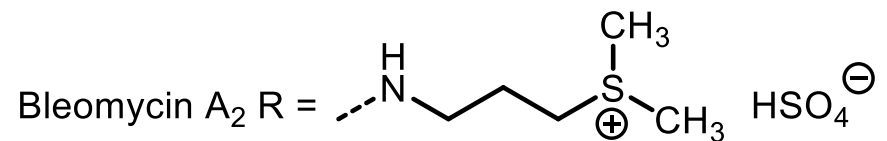
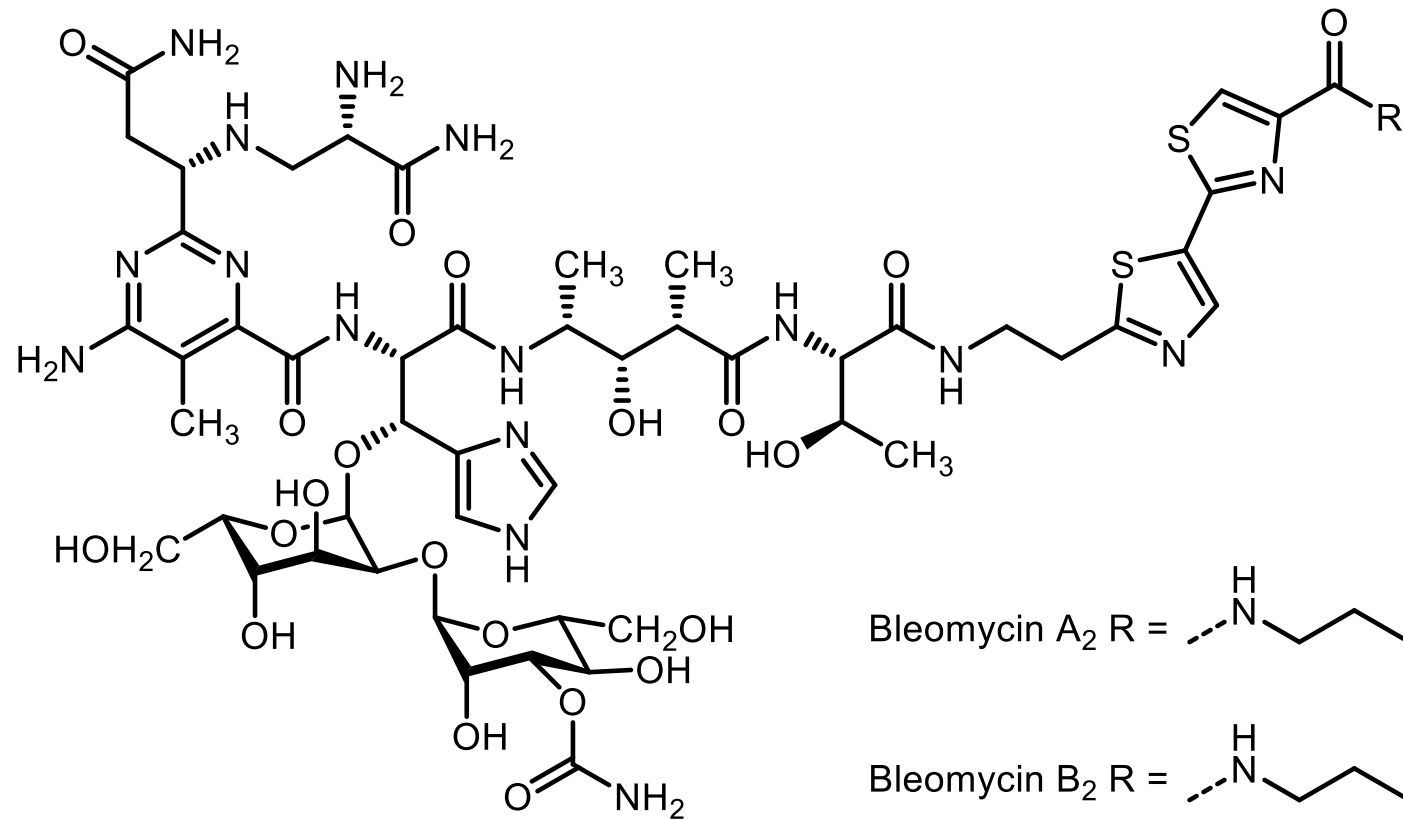
Barbitursäure



Aneurin

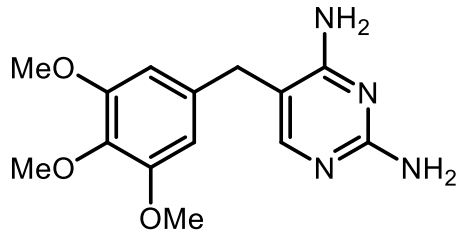
(Thiamin, Vitamin B₁)

5.2.1.1. Pyrimidin Naturstoffe

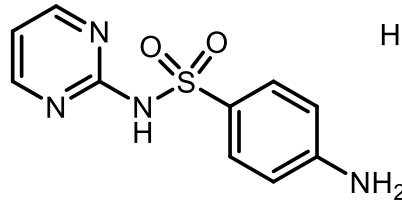


Tumorantibiotika
(von *Streptomyces verticillus* gebildet)

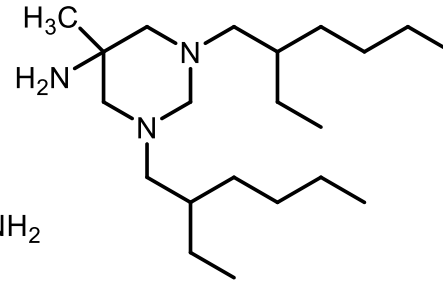
5.2.1.1. Pyrimidin Wirkstoffe



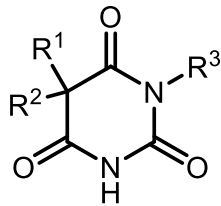
Trimethoprin
(Bakteriostatikum)



Sulfadiazin
(Antiparasitikum)



Hexetidin
(Antiseptikum im Rachenraum)



Barbiturate

$R^1 = R^2 = C_2H_5, R^3 = H$

Veronal, Barbit

$R^1 = C_2H_5, R^2 = Ph, R^3 = H$

Phenobarbital

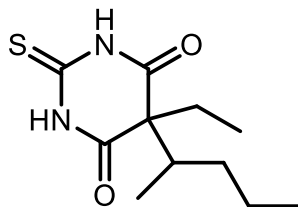
$R^1 = CH_3, R^2 = 1\text{-Cyclohexenyl}, R^3 = CH_3$

Hexobarbital

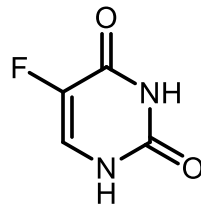
$R^1 = C_2H_5, R^2 = Ph, R^3 = CH_3$

Methylphenobarbital

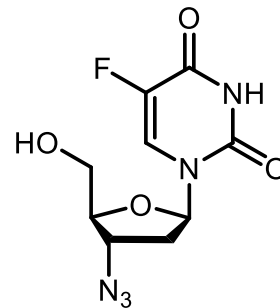
Hypnotica,
Sedativa,
Anxolytica,
Narcotika



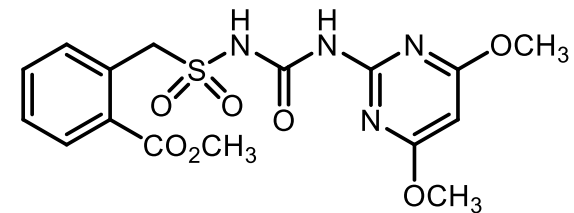
Thiopental
(Kurznarcotikum)



5-Fluoruracil
(Cytostatikum)



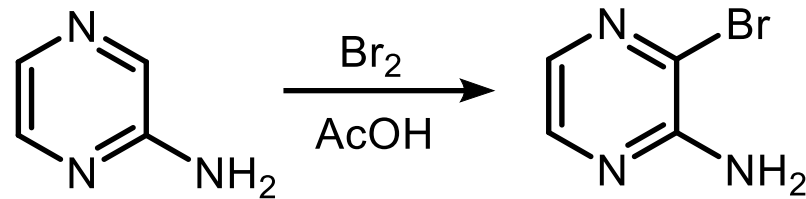
Zidovudin
(antivirales Chemotherapeutikum
bei HIV-Behandlung)



Bensulfuronmethyl
(Herbizid)

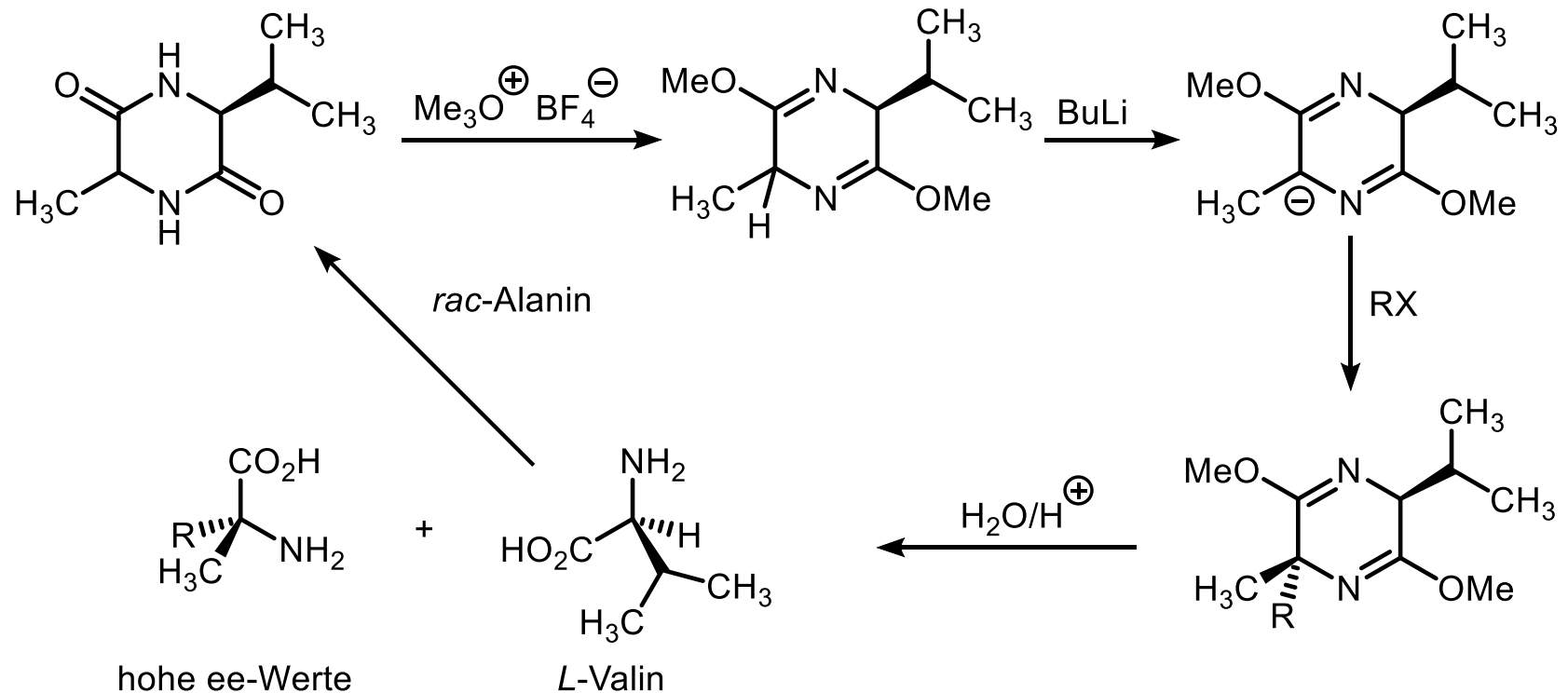
5.2.1.2. Pyrazin

Reaktionen

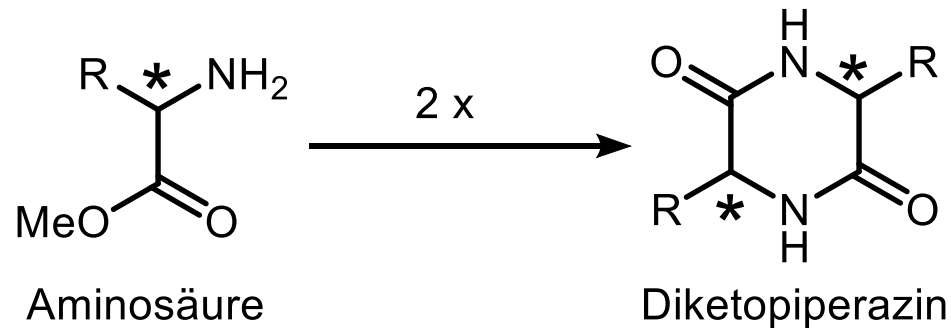
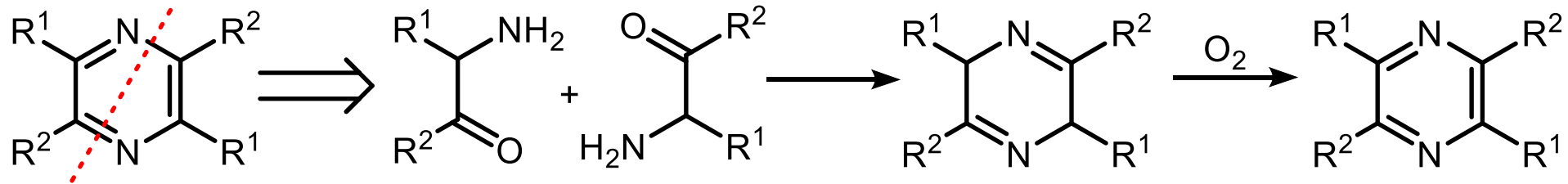
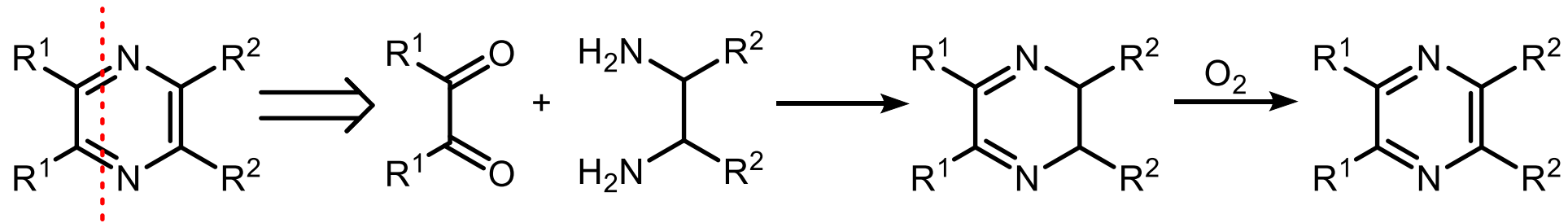


S_E nur bei aktivierten Pyrazinen

Schöllkopf-Bislactimether-Synthesen

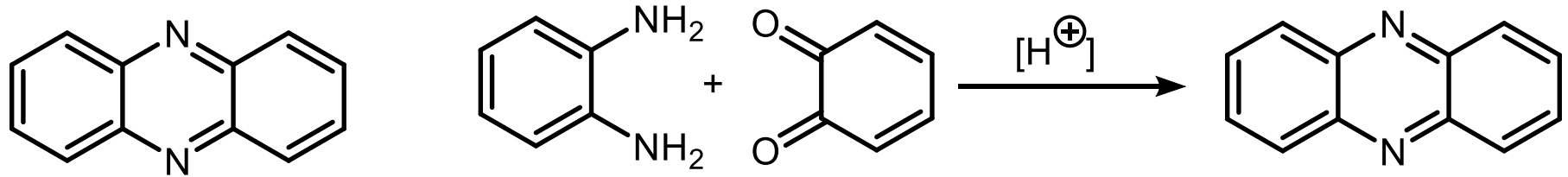


5.2.1.2. Pyrazin Synthesen



5.2.1.2. Pyrazin Benzoanellierte Derivate

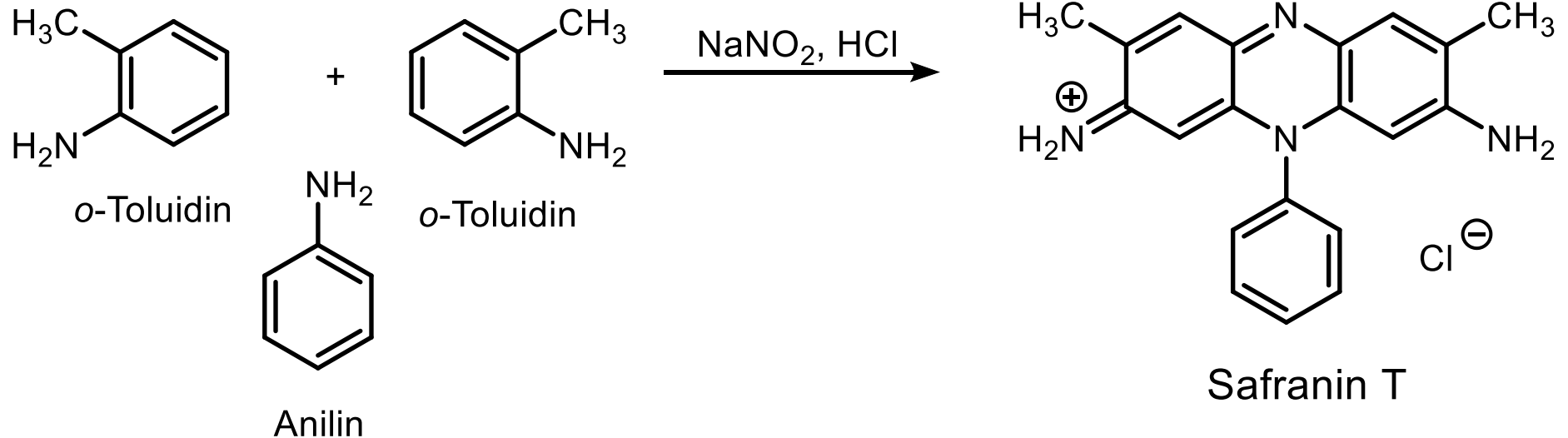
Phenazin



Smp. 171 °C

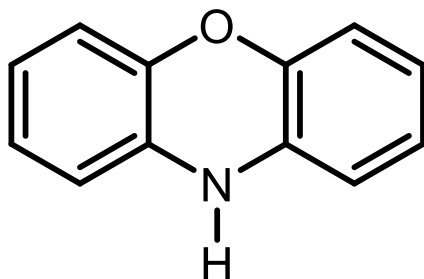
pK_a (Phenazinium) = 1.23

Grundkörper der Safranine
(Klasse: Beizenfarbstoffe)

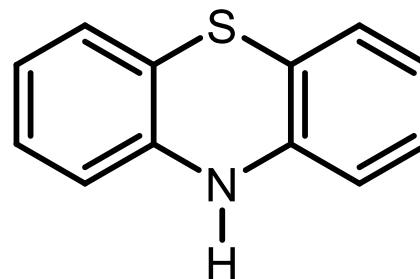


5.2.1.2. Pyrazin

Benzoanellierte Analoga



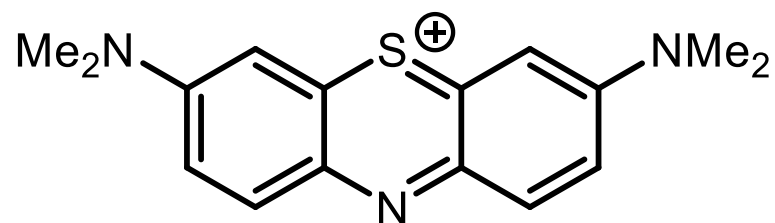
Phenoxazin



Phenothiazin

wegen 16 e^- in den Tricyclen (Hückel-Antiaromat)

=> gewinkelt (Butterfly-Struktur); stabil

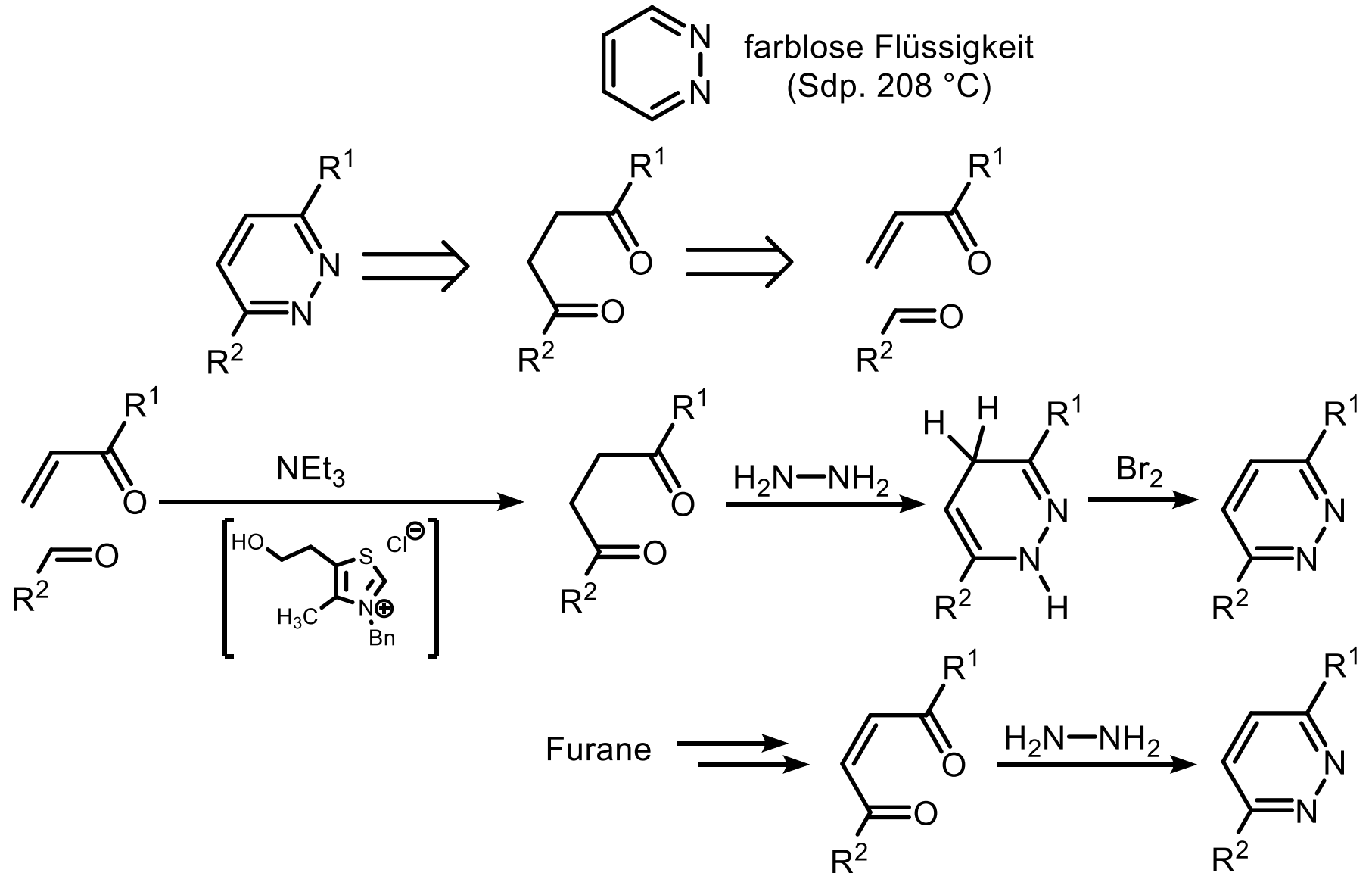


Methylenblau (oxidiertes Phenothiazin)

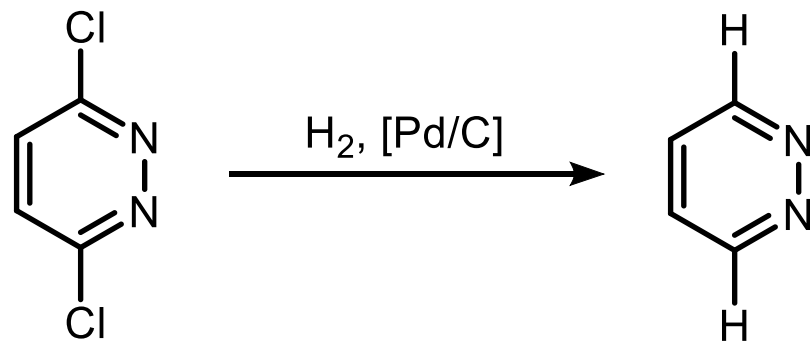
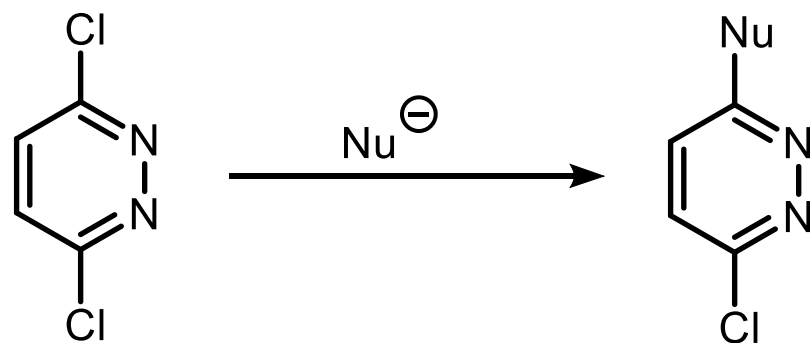
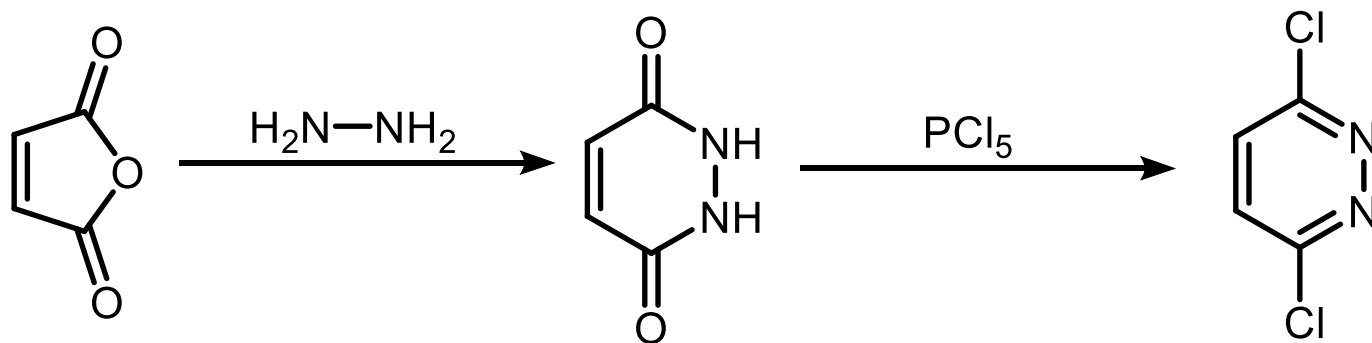
(14 e^- im Tricyclus (Hückel-Aromat))

=> planar; stabil)

5.2.1.3. Pyridazin

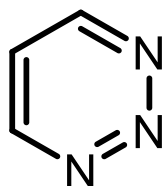


5.2.1.3. Pyridazin

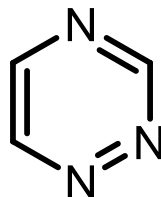


5.3. Sechsring-Heterocyclen mit drei und vier Heteroatomen

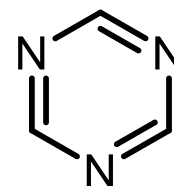
5.3.1. Triazin



1,2,3-Triazin

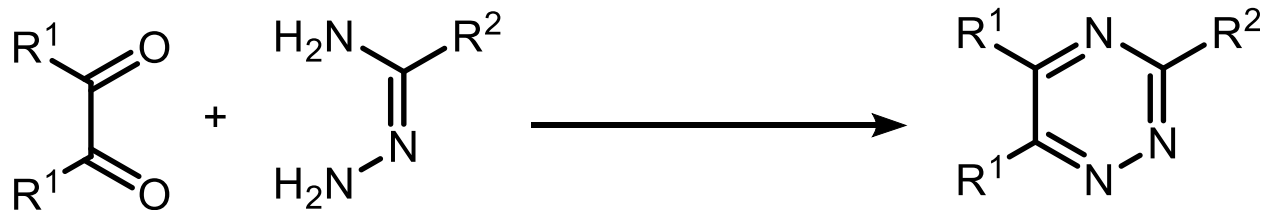


1,2,4-Triazin

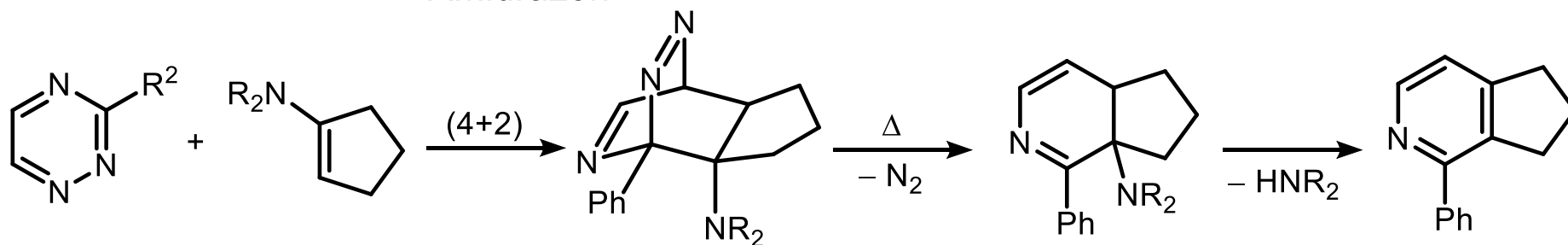


1,3,5-Triazin

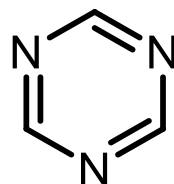
1,2,4-Triazine



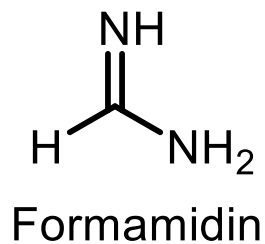
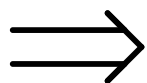
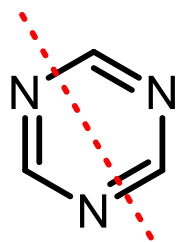
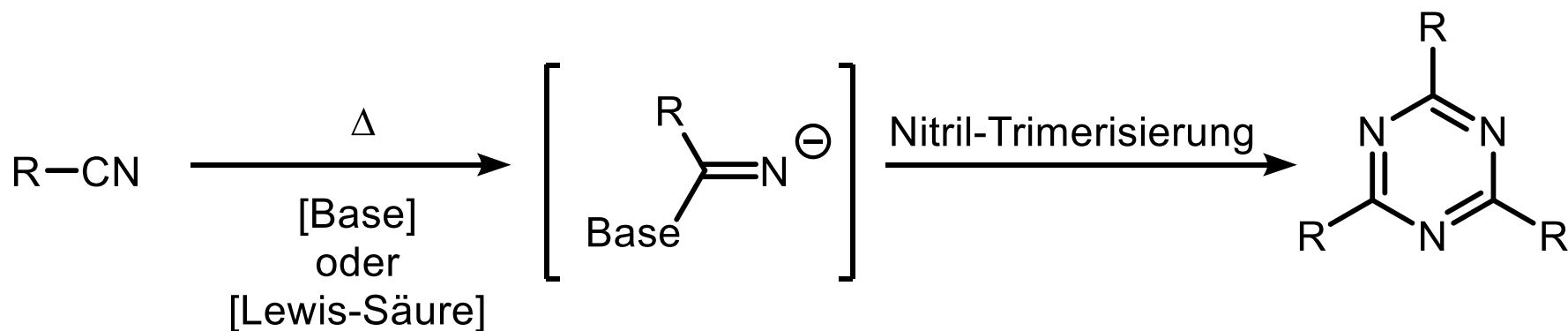
Amidrazon



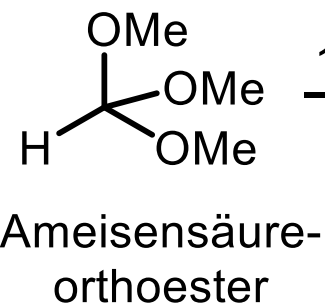
1,3,5-Triazine



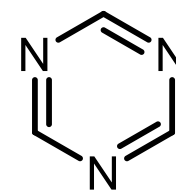
Smp. 80 °C
farblos



+

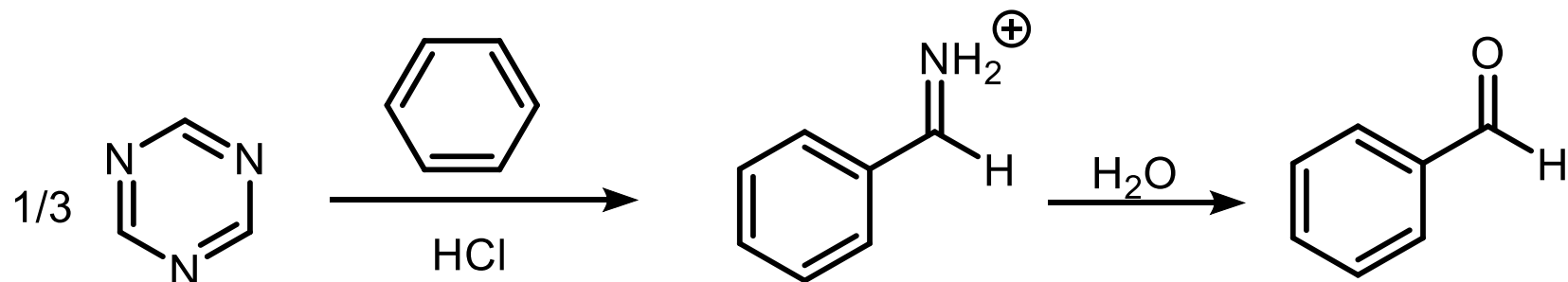


135 °C

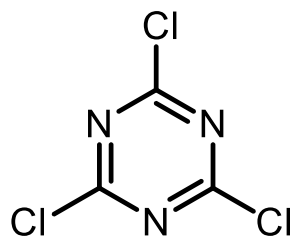


Reaktionen

keine Reaktion mit Elektrophilen; mit Nucleophilen erfolgt Ringöffnung (Formylierung)

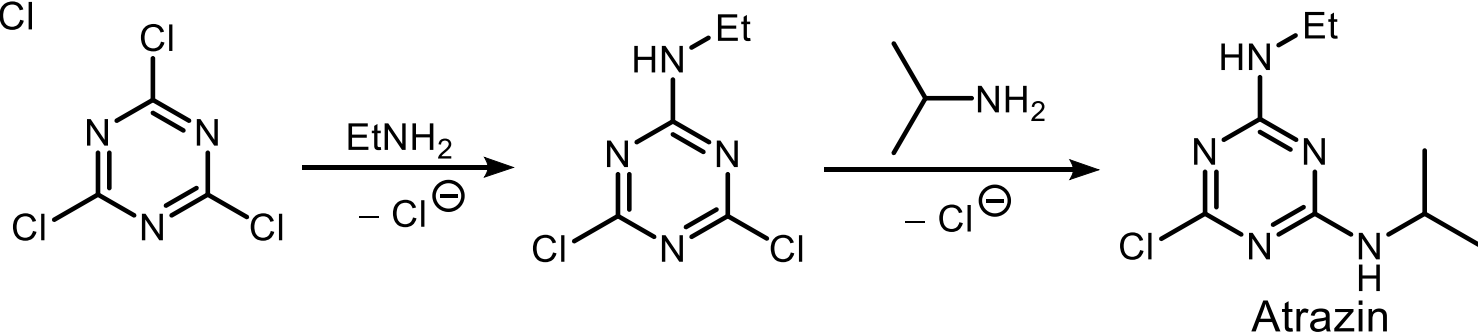


Cyanurchlorid (technische Synthese aus Cl-CN an Aktivkohle)



als Ankergruppe für Farbstoffe, da jede einzelne Cl -Gruppe auch substituiert werden kann;

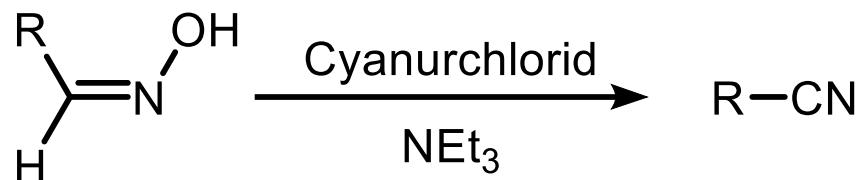
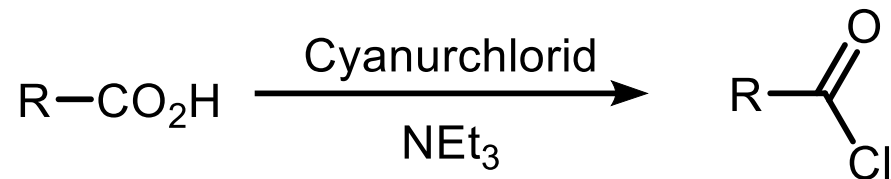
90



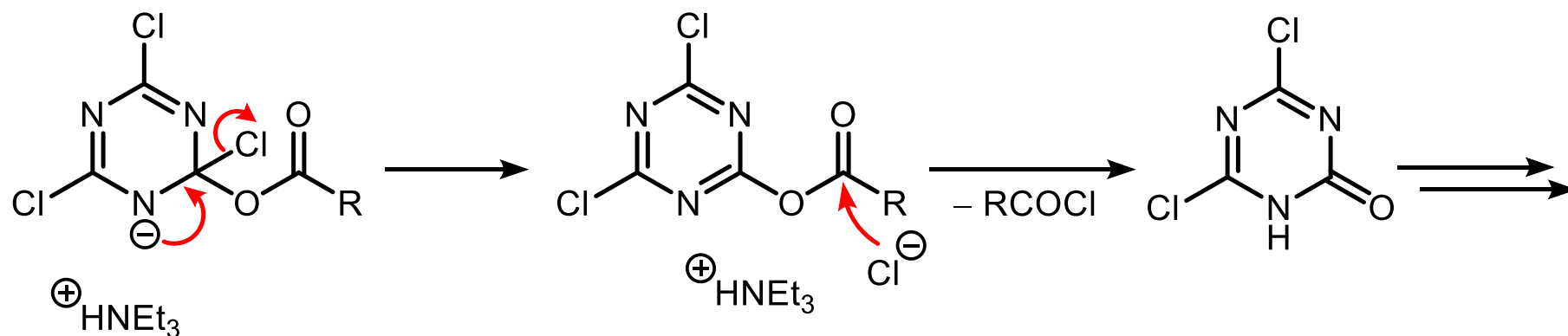
(Herbizid, Photosyntheseinhibitor)

5.3.1. Triazin Cyanurchlorid

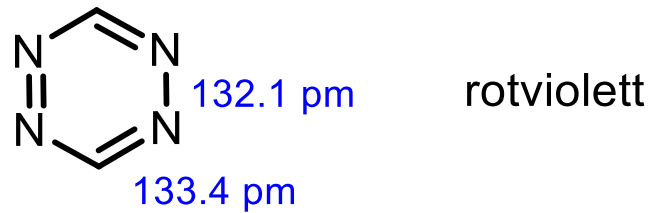
Darstellung von Säurechloriden/Dehydratisierung



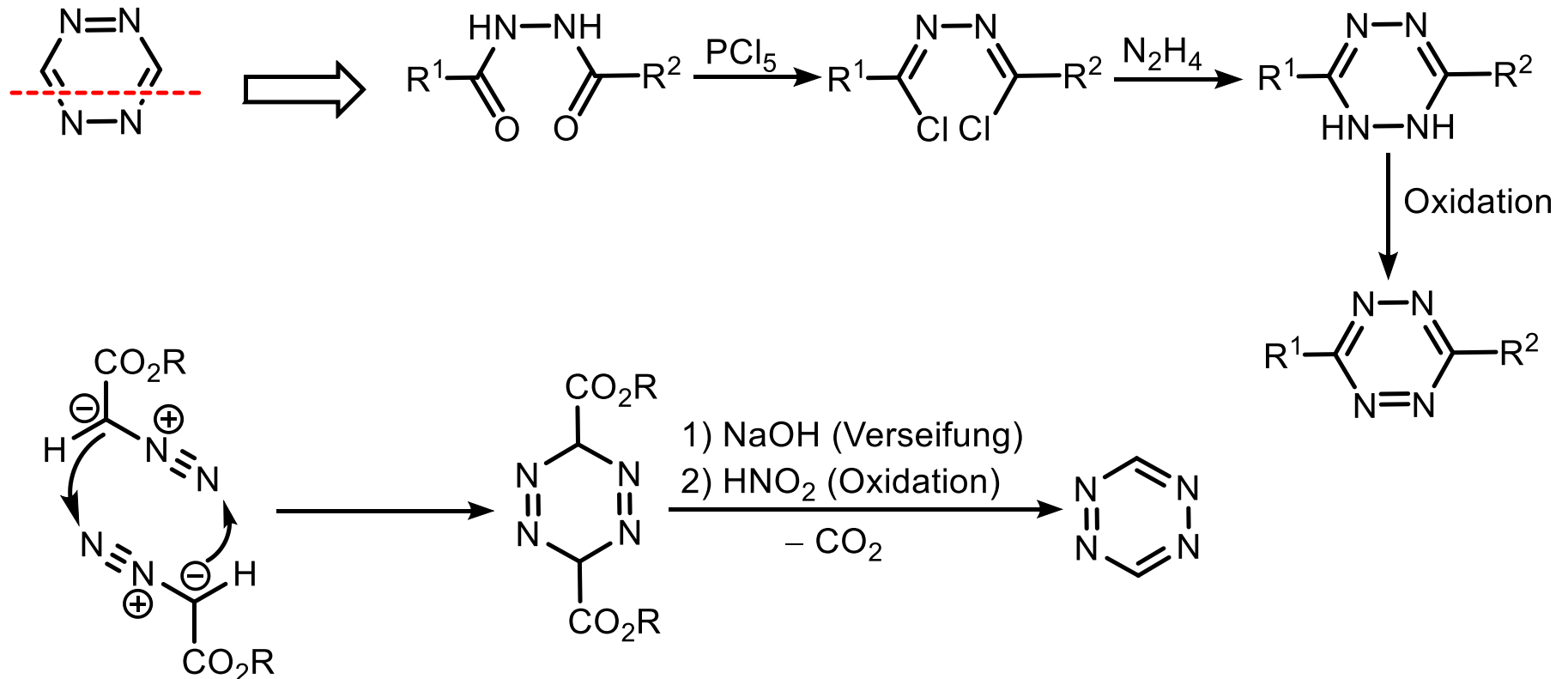
Triebkraft: Bildung von Cyanursäure



5.3.2. 1,2,4,5-Tetrazin



Synthese: Acylierung von Hydrazin



5.3.2. 1,2,4,5-Tetrazin Reaktionen

Diels-Alder-Reaktionen: Synthese von Pyridazinen

